We must distinguish between clues and strong evidence. Clues are what set Sherlock Holmes on the right track, allowing him to solve a mysterious case. Strong evidence is what the judge needs to sentence the guilty. Clues put us on the right path towards a correct theory. Strong evidence is that which subsequently allows us to trust whether the theory we have built is a good one or not. Without clues, we search in the wrong directions. Without evidence, a theory is not reliable.

The same applies to quantum gravity. The theory is in its infancy. Its theoretical apparatus is gaining solidity, and the fundamental ideas are being clarified: the clues are good, and concrete – confirmed predictions are still missing. The theory has not yet taken its exams.

Signals from nature

The most studied alternative to the research direction recounted in this book is string theory. The majority of physicists who have worked on string theory, or string-related theories, expected that as soon as the new particle accelerator at CERN in Geneva began to function (the LHC or Large Hadron Collider), particles of a new kind never before observed, but anticipated by the theory, would immediately become evident: supersymmetric particles. String theory needs these particles to be consistent: that is why the string theorists eagerly expected them to be found. Loop quantum gravity, on the other hand, is well defined even without supersymmetric particles. The loop theorists were inclined to think that these particles might not exist.

The supersymmetric particles were not observed, to the great disappointment of many. The fanfare that greeted the discovery of the Higgs boson in 2013 also masked this disappointment. The supersymmetric particles are not there at the energy where many string theorists expected them to be. This is not a definitive proof of anything – far from it; but nature has given a small clue in favour of the loops.

There have been three major experimental results in fundamental physics in recent years. The first is the revelation of the Higgs boson at CERN in Geneva (figure 9.1). The second is the measurements made by the Planck satellite (figure 9.2), measurements, the data of which were also made public in 2013, confirming the standard cosmological model. The third is the first detection of gravitational waves announced in the first months of 2016. These are the three signals that nature has recently given us.